MA2508 Midterm Tutorial

Multivariable Calculus

Problem 1

If $z = y + f(x^2 - y^2)$, where f is differentiable, show that

$$y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = x.$$

Problem 2

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

- If $\vec{u} \cdot \vec{v} = 0$ and $\vec{u} \times \vec{v} = 0$, then $\vec{u} = 0$ or $\vec{v} = 0$.
- There exists a function f with continuous second-order partial derivatives such that $f_x(x, y) = x - y^2$ and $f_y(x, y) = x + y^2$.

Problem 3

Find the curvature of the parametrized curve

$$\vec{r}(t) = (e^t \cos t, e^t \sin t, e^t), \quad \text{at the point } (1, 0, 1).$$

Problem 4

(a) Find the directional derivative $D_{\vec{u}}f$ of $f = y\cos(x) + y^2$ in direction $\vec{u} = (1, 1)$ at point $(0, 1)$

(b) Find the tangent plane of $z - e^z + 2xy = 3$ at point $(1, 2, 0)$

(c) Given an ellipsoid $x^2 + 2y^2 + 3z^2 = 498$. Find its tangent planes which are parallel to the plane $x + 3y + 5z = 7$.

Problem 5

(a) Consider the following limits. If it exists, find its value. If not, state your reason.

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{2 - e^{xy}} - 1}.$$

(b) Determine whether the following limits exist. If they exist, find their value. If not, show why they do not exist.

$$\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2}.$$

(c) Consider the following limits. If it exists, find its value. If not, state your reason.

$$\lim_{(x,y)\to(0,0)} \frac{xy^3}{x^2 + y^6}.$$

Problem 6

Consider the function

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

(i) Show that $f(x,y)$ is continuous at $(0,0)$.

(ii) Show that the partial derivatives of $f(x,y)$ exist at $(0,0)$.

Problem 7

Let $w = f(u,v)$, where $u(x,y,z) = x^2 + yz$ and $v(x,y,z) = e^{xyz}$, and f has continuous second-order partial derivatives. Compute

$$\frac{\partial w}{\partial y} \qquad \text{and} \qquad \frac{\partial^2 w}{\partial y \partial z}.$$

Problem 8

Let

$$f(x,y) = \begin{cases} 0 & \text{if } y \leq 0 \text{ or } y \geq x^6 \\ 1 & \text{if } 0 < y < x^6 \end{cases}$$

- Show that $f(x,y) \to 0$ as $(x,y) \to (0,0)$ along path through $(0,0)$ of the form $y = mx^a$ with $a < 6$.
- Despite the first part, show that f is discontinuous at $(0,0)$.

Problem 9

Use the method of Lagrange multipliers to find the maximum and minimum values of the function

$$f(x,y,z) = xy + 2z$$

on the sphere

$$x^2 + y^2 + z^2 = 11.$$

Problem 10

Find the maximum and minimum of

$$f(x,y) = x^2 + 2y^2 - x^2y^2$$

in the region

$$D := \{(x,y) \mid x^2 + y^2 \leq 4,\ y \geq 0\}.$$